Time series analysis — exercise sheet 3

AB–204 Arctic population ecology, October 29, 2026

J.-Prof. Dr. Timo Adam

This exercise sheet is about linear regression with trend, seasonal, and cyclical components (Lecture 3).

Before you start: save `sea_ice.csv` (from the course website) in your course folder. At the top of your script, load the colours for categories and the bike sharing data (for Exercise 2):

```
library(viridisLite) # viridis(n, end = 0.9): n colours, one per category
bikes <- read.csv("bike_sharing.csv") # bike sharing data, as on sheets 1 and 2
bikes$t <- 1:nrow(bikes) # time index: 1, 2, ..., 3926
```

Exercise 1: linear regression with trend

1. Use the `read.csv()` function to import the file `sea_ice.csv` (monthly Arctic sea ice extent in million km^2, January 1990 – September 2026), and assign it to the object `ice`.
2. Create a new variable `t` that is the observation index (1, 2, ..., 441).
3. Use the `lm()` function to fit the linear, quadratic, and cubic trend models

$$\begin{aligned}\mathbb{E}(\text{sea ice extent}_t) &= \beta_0 + \beta_1 \cdot t,\\ \mathbb{E}(\text{sea ice extent}_t) &= \beta_0 + \beta_1 \cdot t + \beta_2 \cdot t^2,\\ \mathbb{E}(\text{sea ice extent}_t) &= \beta_0 + \beta_1 \cdot t + \beta_2 \cdot t^2 + \beta_3 \cdot t^3\end{aligned}$$

   to the sea ice data, and assign the fitted models to the objects `mod_1`, `mod_2`, and `mod_3`, respectively.
4. Create a time series plot of the sea ice extent, and add the three fitted trends.
5. Use the `AIC()` function to select between the three trend models.
6. According to the model selected in 1.5, by how much does the expected sea ice extent change per year?
   Hint: the slope $\hat{\beta}_1$ is the change per **month**.
7. The trend models predict the **average** extent over the year. Add the month as a categorical covariate (`as.factor(month)`, see the slides) and predict the sea ice extent in September 2027 and September 2031.
   Hint: September 2027 and September 2031 correspond to $t = 453$ and $t = 501$.

Exercise 2: linear regression with seasonal components

1. Use the `lm()` function to fit the quadratic trend model

$$\mathbb{E}(\text{number of bikes}_t) = \beta_0 + \beta_1 \cdot t + \beta_2 \cdot t^2$$

   to the bike sharing data, and assign the fitted model to the object `mod_5`.
2. Use the `acf()` and `residuals()` functions to plot the autocorrelation function (ACF) of the residuals of `mod_5`, for lags up to 365 days. What do you see?
3. Add a seasonal component to the quadratic trend model from 2.1, i.e., fit the model

$$\mathbb{E}(\text{number of bikes}_t) = \beta_0 + \beta_1 \cdot t + \beta_2 \cdot t^2 + \beta_3 \cdot \sin\left(\frac{2\pi \cdot \text{day of year}_t}{365}\right) + \beta_4 \cdot \cos\left(\frac{2\pi \cdot \text{day of year}_t}{365}\right),$$

   and assign the fitted model to the object `mod_6`.
   Hint: in `R`, π is `pi`.
4. Create a time series plot of the number of bikes, and add the fitted values of `mod_6`.

5. Plot the ACF of the residuals of `mod_6`. What has changed?
6. **Competition:** find a model whose residuals show less autocorrelation, by adding covariates to `mod_6`. The model with the **lowest autocorrelation at lag 1** (closest to 0) wins:

```
acf(residuals(mod_6), plot = FALSE)$acf[2] # 0.62 for mod_6 -- can you do better?
```

Hint: ideas: the day of the week as a categorical covariate, temperature, rain — and the number of bikes on the previous day:

```
bikes$count_lag1 <- c(NA, bikes$count[-nrow(bikes)]) # previous day
```

Exercise 3 (optional): cyclical effects and autoregressive models

1. The data set `lynx` is included in `R`: annual numbers of lynx trapped in north-west Canada, 1821–1934. Create a time series plot, and plot the ACF and the PACF (`pacf()`). What do you see?
2. Use the `arima()` function to fit AR(p) models with $p = 1, 2, 3, 4$, and select between them using the AIC.
   Hint: `AIC()` also works for models fitted with `arima()`.
3. Plot the PACF of the residuals of the selected model. Is there any partial autocorrelation left?